৩ (গ)

ক) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Diff. w. r. to x we get,

$2ax + 2h\left(\frac{dy}{dx}x + y\right) + 2by\frac{dy}{dx} + 2g + 2f\frac{dy}{dx} = 0$

$\Rightarrow 2\frac{dy}{dx}(hx + by + f) = -2(ax + hy + g)$

$\Rightarrow \frac{dy}{dx} = \frac{-(ax + hy + g)}{hx + by + f}$

খ) $x^3 + y^3 - 3axy = 0$

Diff. w. r. to x we get,

$3x^2 + 3y^2\frac{dy}{dx} - 3a\left(y + x\frac{dy}{dx}\right) = 0$

$\Rightarrow 3\frac{dy}{dx}(y^2 - ax) = 3(ay - x^2)$

$\Rightarrow \frac{dy}{dx} = \frac{ay - x^2}{y^2 - ax}$

গ) $x^{2/3} + y^{2/3} = a^{2/3}$

Diff. w. r. to x we get,

$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0$ বা, $\frac{dy}{dx} = -\left(\frac{y}{x}\right)^{1/3}$

$\Rightarrow y^{-1/3}\frac{dy}{dx} = -x^{-1/3}$

$\rightarrow \frac{dy}{dx} = -\left(\frac{x}{y}\right)^{-1/3}$

8] $x^n + y^n = a^n$

Diff. w. r. to x we get,

$$n x^{n-1} + n y^{n-1} \frac{dy}{dx} = 0$$

$$y^{n-1} \frac{dy}{dx} = -x^{n-1}$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{x}{y}\right)^{n-1}$$

9] $\ln(xy) = x^2 + y^2$

Diff. w. r. to x we get,

$$\frac{1}{xy}\left(y + x\frac{dy}{dx}\right) = 2x + 2y\frac{dy}{dx}$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y}\frac{dy}{dx} - 2y\frac{dy}{dx} = 2x$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{1}{y} - 2y\right) = 2x - \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{1-2y^2}{y}\right) = \frac{2x^2-1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y(2x^2-1)}{x(1-2y^2)}$$

8) $x^2+y^2=\sin(xy)$

Diff. w. r. to x we get,

$$2x+2y\frac{dy}{dx}=\cos(xy)\left(x\frac{dy}{dx}+y\right)$$

$$\Rightarrow 2y\frac{dy}{dx}-x\cos(xy)\frac{dy}{dx}=y\cos(xy)-2x$$

$$\Rightarrow \frac{dy}{dx}=\frac{y\cos(xy)-2x}{2y-x\cos(xy)}$$

9) $x^2y+y^2x+\sqrt{xy}=1$

Diff. w. r. to x we get,

$$2xy+x^2\frac{dy}{dx}+2xy\frac{dy}{dx}+y^2+\frac{1}{2\sqrt{xy}}\left(y+x\frac{dy}{dx}\right)=0$$

$$\Rightarrow \frac{dy}{dx}\left(x^2+2xy+\frac{x}{2\sqrt{xy}}\right)=-\left(y^2+2xy+\frac{y}{2\sqrt{xy}}\right)$$

$$\Rightarrow \frac{dy}{dx}=\frac{-\left(y^2+2xy+\frac{\sqrt{y}}{2\sqrt{x}}\right)}{x^2+2xy+\frac{\sqrt{x}}{2\sqrt{y}}}$$

৯। $x\sqrt{1-y^2} + y\sqrt{1-x^2} = a$

Diff. w. r. to x we get,

$$\frac{x}{2\sqrt{1-y^2}}\left(-2y\frac{dy}{dx}\right) + \sqrt{1-y^2} + \frac{y}{2\sqrt{1-x^2}}(-2x) + \sqrt{1-x^2}\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx}\left(\sqrt{1-x^2} - \frac{xy}{\sqrt{1-y^2}}\right) = \frac{xy}{\sqrt{1-x^2}} - \sqrt{1-y^2}$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{\sqrt{1-x^2}\cdot\sqrt{1-y^2} - xy}{\sqrt{1-y^2}}\right) = -\frac{(\sqrt{1-x^2}\cdot\sqrt{1-y^2} - xy)}{\sqrt{1-x^2}}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

২০। $\sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\} = c$

$\Rightarrow \sin^{-1}x + \sin^{-1}y = c$

Diff. w. r. to x we get,

$$\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

22) $y = x \ln y$

Diff w.r. to x we get,

$$\frac{dy}{dx} = \ln y + \frac{x}{y}\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}\left(1 - \frac{x}{y}\right) = \ln y$$

$$\Rightarrow \frac{dy}{dx}\left(\frac{y-x}{y}\right) = \ln y$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \ln y}{y-x}$$

22) $\sin(x+y) = a$

Diff. w.r. to x we get,

$$\cos(x+y)\left(1 + \frac{dy}{dx}\right) = 0$$

$$\Rightarrow \cos(x+y)\frac{dy}{dx} = -\cos(x+y)$$

$$\Rightarrow \frac{dy}{dx} = -1$$

20| $\sin y = x\sin(a+y)$

Diff. w. r. to x we get,

$$\cos y\frac{dy}{dx} = \sin(a+y) + x\cos(a+y)\left(0+\frac{dy}{dx}\right)$$

$$\Rightarrow \cos y\frac{dy}{dx} - x\cos(a+y)\frac{dy}{dx} = \sin(a+y)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin(a+y)}{\cos y - x\cos(a+y)}$$

28| $x\cos y = \sin(x+y)$

Diff. w. r. to x we get,

~~$\cos y \frac{dy}{dx} +$~~

$$\cos y - x\sin y\frac{dy}{dx} = \cos(x+y)\left(1+\frac{dy}{dx}\right)$$

$$\Rightarrow \frac{dy}{dx}\left(\cos(x+y) + x\sin y\right) = -\cos(x+y) + \cos y$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\cos(x+y) + \cos y}{\cos(x+y) + x\sin y}$$

২৩। $y = \sin(x+y)^2$

$$\frac{dy}{dx} = \cos(x+y)^2 \left\{2(x+y)\left(1+\frac{dy}{dx}\right)\right\}$$

$$\Rightarrow \frac{dy}{dx} = 2(x+y)\cos(x+y)^2 + 2(x+y)\left(\cos(x+y)^2\right)\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}\left(1-2(x+y)\cos(x+y)^2\right) = 2(x+y)\cos(x+y)^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{2(x+y)\cos(x+y)^2}{1-2(x+y)\cos(x+y)^2}$$

Sol $\sqrt{\frac{y}{x}} + \sqrt{\frac{x}{y}} = 3$

$\Rightarrow \frac{y+x}{\sqrt{xy}} = 3$

$\Rightarrow x+y = 3\sqrt{xy}$

$\Rightarrow x^2+2xy+y^2-6xy=0$

$\Rightarrow x^2-4xy+y^2=0$

Diff. w.r. to x we get,

$2x-4\left(y+x\frac{dy}{dx}\right)+2y\frac{dy}{dx}=0$

$\Rightarrow 2x - 4y - 4x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx}(y-2x) = 2y - x$

$\Rightarrow \frac{dy}{dx} = \frac{2y-x}{y-2x}$

2a) $\sqrt{x/y} + \sqrt{y/x} = a$

$\Rightarrow x + y = a\sqrt{xy}$

$\Rightarrow x^2 + 2xy + y^2 = a^2xy$

Diff. w.r. to x we get,

$2x + (2-a^2)\left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx}(2x - a^2x + 2y) = -\{(2-a^2)y + 2x\}$

$\Rightarrow \frac{dy}{dx} = \frac{-\{(2-a^2)y + 2x\}}{2y + (2-a^2)x}$

2b) $\sqrt{x/y} + \sqrt{y/x} = 1$

$\Rightarrow x + y = \sqrt{xy}$

$\Rightarrow x^2 + 2xy + y^2 = xy$

$\Rightarrow x^2 + xy + y^2 = 0$

Diff. w.r. to x we get

$2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$

$\Rightarrow \frac{dy}{dx}(x+2y) = -(2x+y)$

$\Rightarrow \frac{dy}{dx} = \frac{-(2x+y)}{2y+x}$

১৯। $x^y = y^x$

$y \ln x = x \ln y$

$\Rightarrow \dfrac{y}{x} + \ln x \dfrac{dy}{dx} = \dfrac{x}{y}\dfrac{dy}{dx} + \ln y.$

$\Rightarrow \dfrac{dy}{dx}\left(\ln x - \dfrac{x}{y}\right) = \ln y - \dfrac{y}{x}$

$\Rightarrow \dfrac{dy}{dx}\left(\dfrac{y\ln x - x}{y}\right) = \dfrac{x \ln y - y}{x}$

$\Rightarrow \dfrac{dy}{dx} = \cancel{x(y\ln x - x)} \quad \dfrac{y(x\ln y - y)}{x(y\ln x - x)}$

২০। $(x+y)^{m+n} = x^m y^n$

$(m+n)\ln(x+y) = m\ln x + n \ln y$

Diff. w. r. to x we get,

$\dfrac{m+n}{x+y}\left(1 + \dfrac{dy}{dx}\right) = \dfrac{m}{x} + \dfrac{n}{y}\dfrac{dy}{dx}$

$\Rightarrow \dfrac{dy}{dx}\left(\dfrac{m+n}{x+y} - \dfrac{n}{y}\right) = \dfrac{m}{x} - \dfrac{m+n}{x+y}$

$\Rightarrow \dfrac{dy}{dx} \dfrac{y(m+n) - n(x+y)}{y(x+y)} = \dfrac{m(x+y) - x(m+n)}{x(x+y)}$

$\Rightarrow \dfrac{dy}{dx} \dfrac{my - nx}{y(x+y)} = \dfrac{my - nx}{x(x+y)}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{y}{x}$

22| $x^y + y^x = a^b$

Diff. w. r. to x we get.

$$x^y\left[y\frac{d}{dx}\ln x + \ln x\frac{dy}{dx}\right] + y^x\left[x\frac{d}{dx}\ln y + \ln y\right] = 0$$

$$\Rightarrow x^y\left[\frac{y}{x} + \ln x\frac{dy}{dx}\right] + y^x\left[\frac{x}{y}\frac{dy}{dx} + \ln y\right] = 0$$

$$\Rightarrow yx^{y-1} + x^y\ln x\frac{dy}{dx} + xy^{x-1}\frac{dy}{dx} + y^x\ln y = 0$$

$$\Rightarrow \frac{dy}{dx}\left(x^y\ln x + xy^{x-1}\right) = -\left(y^x\ln y + yx^{y-1}\right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{-\left(y^x\ln y + yx^{y-1}\right)}{x^y\ln x + xy^{x-1}}$$

22) $x^y + y^x = (x+y)^{x+y}$

$\Rightarrow u + v = z$

$\Rightarrow \dfrac{du}{dx} + \dfrac{dv}{dx} = \dfrac{dz}{dx}$ — (1)

$u = x^y$

$\ln u = y \ln x$

$\dfrac{1}{u}\dfrac{du}{dx} = \dfrac{y}{x} + \ln x \dfrac{dy}{dx}$

$\dfrac{du}{dx} = u\dfrac{y}{x} + u \ln x \dfrac{dy}{dx}$

$= y x^{y-1} + x^y \ln x \dfrac{dy}{dx}$

$v = y^x$

$\ln v = x \ln y$

$\dfrac{1}{v}\dfrac{dv}{dx} = \dfrac{x}{y}\dfrac{dy}{dx} + \ln y$

$\Rightarrow \dfrac{dv}{dx} = v\dfrac{x}{y}\dfrac{dy}{dx} + v \ln y$

$= x y^{x-1}\dfrac{dy}{dx} + y^x \ln y$

$z = (x+y)^{x+y}$

$\ln z = (x+y)\ln(x+y)$

$\dfrac{1}{z}\dfrac{dz}{dx} = \dfrac{x+y}{x+y}\left(1 + \dfrac{dy}{dx}\right)$

$\dfrac{dz}{dx} = (x+y)^{x+y} + (x+y)^{x+y}\dfrac{dy}{dx}.$

20| $y = \sin^{-1}\left(\frac{ax}{y}\right)$

$$\cancel{\frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{a^2x^2}{y^2}}}}$$

$$\sin y = \frac{ax}{y}$$

$\Rightarrow$ $y \sin y = ax$

Diff. w. r. to x we get.

$$y\cos y \frac{dy}{dx} + \sin y \frac{dy}{dx} = a$$

$$\Rightarrow \frac{dy}{dx} = \frac{a}{y\cos y + \sin y}$$

28| $x+y = \sin^{-1}\frac{y}{x}$

$\Rightarrow$ $x\sin(x+y) = y$

Diff. w. r. to x we get.

$$\sin(x+y) + x\cos(x+y)\left(1+\frac{dy}{dx}\right) = \frac{dy}{dx}.$$

$$\Rightarrow \frac{dy}{dx}\left(1 - x\cos(x+y)\right) = \sin(x+y) + x\cos(x+y)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin(x+y) + x\cos(x+y)}{1 - x\cos(x+y)}$$

23) $x = a(t + \sin t)$

$\frac{dx}{dt} = a(1 + \cos t)$

$y = a(1 - \cos t)$

$\frac{dy}{dt} = a(\sin t)$

$$\frac{dy}{dx} = \frac{a \sin t}{a(1+\cos t)} = \frac{2\sin t/2 \cos t/2}{2\cos^2 t/2} = \tan(t/2)$$

24) $x = a(\cos t + t \sin t)$

$y = a(\sin t - t\cos t)$

$\frac{dx}{dt} = a(-\sin t + t\cos t + \sin t)$

$\frac{dy}{dt} = a(\cos t + t\sin t - \cos t)$

$$\therefore \frac{dy}{dx} = \frac{\cancel{\cos t}\ t\sin t}{t\cos t} = \tan t$$

29) $x = \sin^2 t$

$y = \tan t$

$\frac{dx}{dt} = 2\sin t \cos t$

$= \sin 2t$

$\frac{dy}{dt} = \sec^2 t$

$$\therefore \frac{dy}{dx} = \sec^2 t \operatorname{cosec} 2t$$

2⑥ $\cancel{x = a\cos}$ $x = \sin^{-1}\frac{2t}{1+t^2}$

$= 2\tan^{-1} t$

$\therefore \frac{dx}{dt} = \frac{2}{1+t^2}$

$y = \tan^{-1}\frac{2t}{1-t^2}$

$y = 2\tan^{-1} t$

$\frac{dy}{dt} = \frac{2}{1+t^2}$

$$\therefore \frac{dy}{dx} = 1$$

২৮। $x = a\cos^3 t$ $\quad$ $y = a\sin^3 t$

$\frac{dx}{dt} = -3a\cos^2 t\ \sin t$ $\quad$ $\frac{dy}{dt} = 3a\sin^2 t\ \cos t$

$\cancel{\frac{dy}{dx} = \cot^2 t \cdot \tan t = \cot t}$

$\therefore \frac{dy}{dx} = \cancel{\cos} - \frac{\sin^2 t\cos t}{\cos^2 t\ \sin t} = -\tan t$

২৯। $x = a\cos^3 e^t$ $\quad$ $y = a\sin^3 e^t$

$\frac{dx}{dt} = -3a\cos^2 e^t\ \sin e^t\ e^t$ $\quad$ $\frac{dy}{dt} = 3a\sin^2 e^t\ \cos e^t\ e^t$

$\therefore \frac{dy}{dx} = \tan e^t$

৩০। $x = 3\tan^{-1}\frac{2t}{1-t^2}$ $\quad$ $y = \cancel{\tan^{-1}}\ \sin^{-1}\frac{2t}{1+t^2}$

$\Rightarrow x = 3\times 2\tan^{-1} t$ $\quad$ $y = 2\tan^{-1} t$

$\therefore \frac{dx}{dt} = \frac{6}{1+t^2}$ $\quad$ $\frac{dy}{dt} = \frac{2}{1+t^2}$

$\therefore \frac{dy}{dx} = \frac{1}{3}$

৩৪। $x = \dfrac{3at}{1+t^3}$

$$\frac{dx}{dt} = \frac{(1+t^3)3a - 3at \cdot 3t^2}{(1+t^3)^2} = \frac{3a+3at^3-9at^3}{(1+t^3)^2}$$

$$= \frac{3(a-2at^3)}{(1+t^3)^2}$$

$y = \dfrac{3at^2}{1+t^3}$

$$\frac{dy}{dt} = \frac{(1+t^3)6at - 3at^2 \cdot 3t^2}{(1+t^3)^2} = \frac{6at+6at^4-9at^4}{(1+t^3)^2}$$

$$= \frac{3t(2a-at^3)}{(1+t^3)^2}$$

$$\therefore \frac{dy}{dx} = \frac{2at-at^4}{a-2at^3} = \frac{2t-t^4}{1-2t^3}$$

৩৫। $x = \dfrac{\sin^3 t}{\sqrt{\cos 2t}}$

$$\frac{dx}{dt} = \frac{\sqrt{\cos 2t}\, 3\sin^2 t \cos t + \sin^3 t \cdot \frac{1}{2\sqrt{\cos 2t}} 2\sin 2t}{\cos 2t}$$

$$= \frac{3\sin^2 t \cdot \cos t \cdot \cos 2t + \sin^3 t \cdot \sin 2t}{(\sqrt{\cos 2t})^3}$$

$$y = \frac{\cos^3 t}{\sqrt{\cos 2t}}$$

$$\frac{dy}{dt} = \frac{-\sqrt{\cos 2t}\cdot 3\cos^2 t\,\sin t + \cos^3 t\cdot \frac{1}{2\sqrt{\cos 2t}}\,2\sin 2t}{\cos 2t}$$

$$= \frac{\cos^3 t\cdot \sin 2t - 3\cos^2 t\,\sin t\cdot \cos 2t}{\left(\sqrt{\cos 2t}\right)^3}$$

$$\therefore \frac{dy}{dx} = \frac{\cos^3 t\cdot \sin 2t - 3\cos^2 t\,\sin t\cdot \cos 2t}{3\sin^2 t\cdot \cos t\cdot \cos 2t + \sin^3 t\cdot \sin 2t}$$